Stat 360 - Review in Stat 495

For our STAT 360 review, we'll tackle matching and then a few problems to revisit some typical computations in probability theory.

Matching Activity

For each description below, match the most appropriate selection (using clear capital letters). Not all options may be used.

Description	Match Options
________ Min and Max are examples	A. Markov Chains
________ Building block distribution for discrete dists	B. Support
________ Came up with the axioms	C. Bivariate Xform
________ Convergence in probability result	D. Bernoulli
________ Convergence in distribution result	E. Gamma
________ Scaled version of covariance	F. Inverse Transform
________ Region over which pdfs take + probability	G. Kolmogorov
________ Used to model trial number of first success	H. CLT
________ Right-skewed distribution with two parameters	I. Binomial
________ Requires the Jacobian to perform	J. Geometric
	K. WLLN
	L. SLLN
	M. Order statistics
	N. Correlation
	O. Exponential

Exercises

For the exercises below, assume that X is a continuous RV with pdf $f(x) = 2x, 0 < x < 1$, and 0, otherwise.

Find the CDF of X and fully specify it.

Find $E(X)$ via computation.

Identify the distribution of X as one you know and use its known properties to verify your $E(X)$ computation.

Be sure you could do the variance computation and verification as well (do it out if you're not sure!).

Find $E(3X+1)$. Remember there are lots of rules about including scalars in these computations!

In theoretical statistics, our focus is often on the parameters of the distribution. What are the distributional parameters for this distribution, in general? Remember that the most general notation we have for a parameter is usually θ, depending on your text.

Find $P(0.3 < X < 0.6)$ by hand computation.

Find $P(0.3 < X < 0.6)$ using appropriate R commands to verify your computation. Write down the command you used.

Now suppose $Y = 4X$. Find and fully specify the pdf of Y.

Room to continue finding the pdf.

Suppose $X_1, \cdots, X_n$ are i.i.d. observations with the same distribution as X. What does i.i.d. mean / stand for?

Find the joint pdf of $X_1, \cdots, X_n$. How do you know you can do this? What notation do you need?

Suppose W is a RV with $E(W) = 1, V(W) = 1$, and $cor(X, W) = 0.6$. What is the covariance between X and W? Are X and W independent? Explain.